Week 1

MICROECONOMIC THEORY
ECON 323 502/503

Spring 2026

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Chapter 3. Consumer Behavior -- 1

Corresponding to the following chapters in the textbook (Microeconomics, Pindyck and Rubinfeld)

- Chapter 3.1 Consumer Preferences
- Chapter 3.2 Budget Constraints
- Chapter 3.3 Consumer Choice
- Chapter 3.5 Marginal Utility and Consumer Choice

Chapter 3. Consumer Behavior -- 2

What do we study in this chapter?

- How a consumer with limited income decides which goods and services to buy.
- This helps us understand where the demand function comes from.

Steps:

- Consumer Preferences (What do you like more? Intrinsic. Not related to prices/income)
- Budget Constraint (Decided by prices and income)
- Consumer Choice (How to make decisions given preferences and constraint?)

Chapter 3.1 Consumer Preferences -- 1

- **Market basket/bundle** is a list with specific quantities of all goods.

- E.g., every month, Evan consumes (90 meals, 20 gallons of gas, 3000 gallons of water, 1 piece of clothing, …) -> market basket

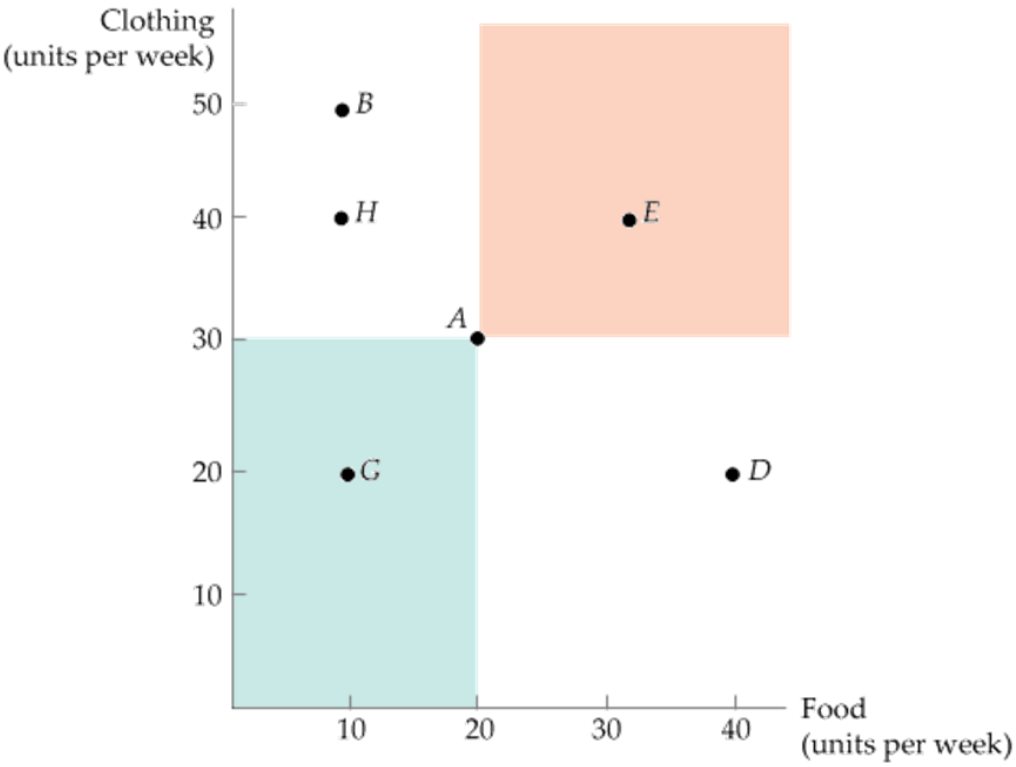
- For simplicity, we usually assume in this class that there are only two goods in the economy.

- Write a market basket of x units of Good X and y units of Good Y as (x, y) and we can plot (x, y) on x-y two-axis plane.

Chapter 3.1 Consumer Preferences -- 2

Examples of six market baskets in an economy with food and clothing:

MARKET BASKET	UNITS OF FOOD	UNITS OF CLOTHING
A	20	30
B	10	50
D	40	20
E	30	40
G	10	20
H	10	40



Chapter 3.1 Consumer Preferences -- 3

How would a consumer compare these market baskets?

This is described by **preferences**.

Preferences are intrinsic. They have nothing to do with price of goods or the consumers' income level.

In principle, a consumer can have any kind of preferences, but for analytical reasons, we impose some (often) reasonable assumptions on preferences.

Basic Assumptions about Preferences

Preferences are assumed to be *complete*.

Assumption 1. Completeness: **Consumers can compare and rank all possible baskets.**

- Thus, for any two market baskets *A* and *B*, a consumer will prefer *A* to *B*, will prefer *B* to *A*, or will be indifferent between the two.

- Example: What do you prefer to receive as a birthday gift? A PlayStation or an iPad? If you cannot decide (not because they are indifferent), your preference violates completeness.

Chapter 3.1 Consumer Preferences -- 5

Basic Assumptions about Preferences

Preferences are assumed to be *transitive*.

Assumption 2. Transitivity: **If a consumer prefers basket *A* to basket *B* and basket *B* to basket *C*, then the consumer also prefers *A* to *C*.**

- Example: If you prefer Honda Accord to Toyota Camry, prefer Toyota Camry to Ford Fusion, and prefer Ford Fusion to Honda Accord simultaneously, then your preference violates transitivity.

Basic Assumptions about Preferences

Preferences are assumed to satisfy more is better than less.

Assumption 3. More is better than less: ***Consumers always prefer more of any good to less*.**

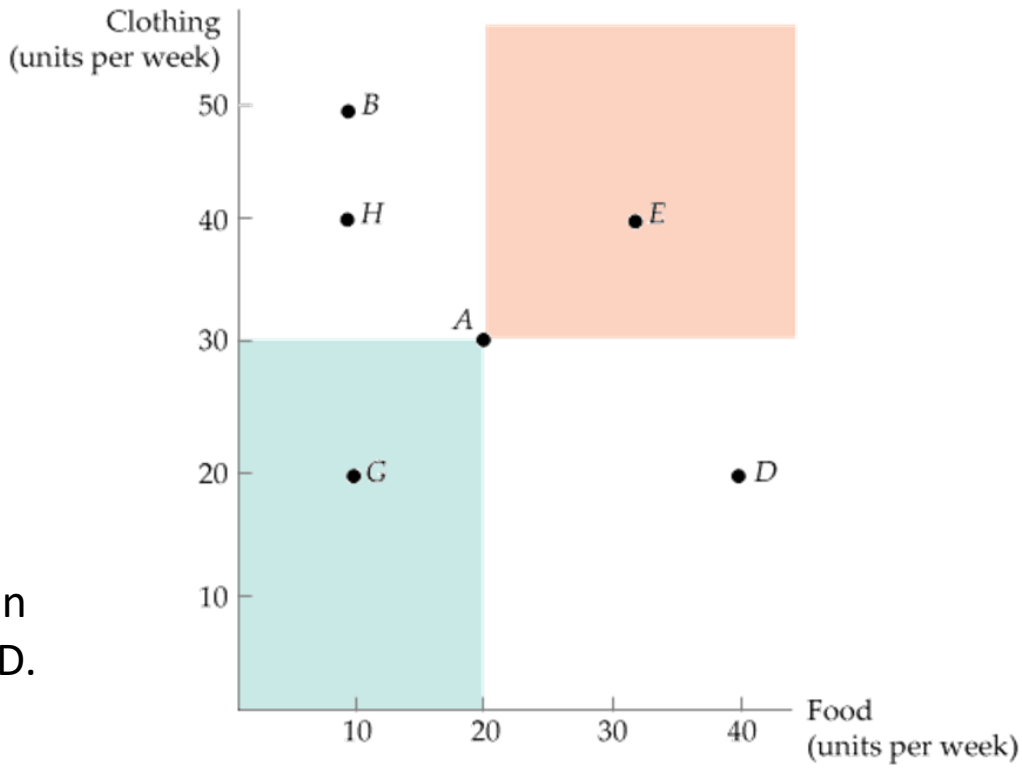
- Goods are assumed to be desirable—i.e., to be *good.*
- Consumers are never satiated.
- Example: In a two-good economy, if Evan prefers (2 units of food, 1 unit of clothing) to (3 units of food, 2 units of clothing), then his preference violates the more is better than less assumption.

Chapter 3.1 Consumer Preferences -- 7

Suppose the three assumptions hold.

Let's compare A with other market baskets.

- More is better -> E is preferred to A;
- More is better -> A is preferred to G;
- Need further information to compare A with B/H/D.



Indifference Curves

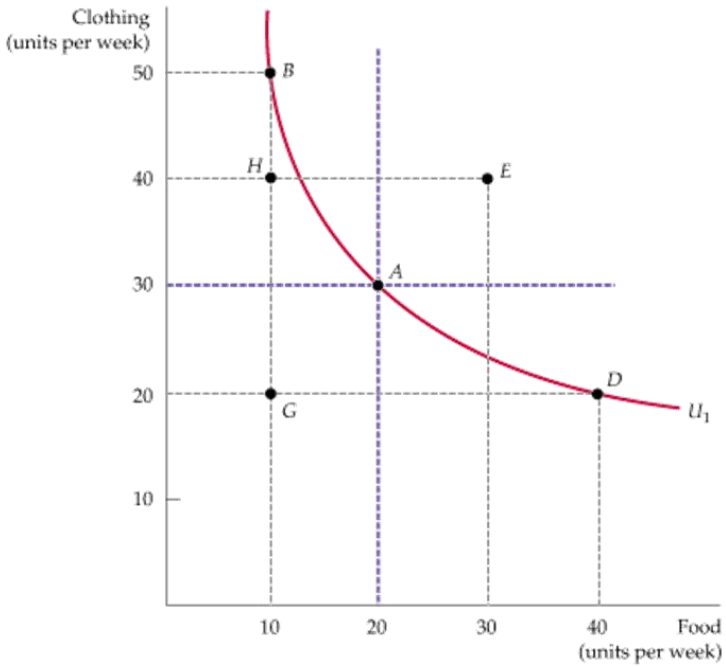
Indifference Curve: Curve representing all combinations of market baskets that provide a consumer with the same level of satisfaction.

According to this indifference curve:

- B is indifferent to A
- D is indifferent to A

Combine with the assumptions of more is better than less and transitivity

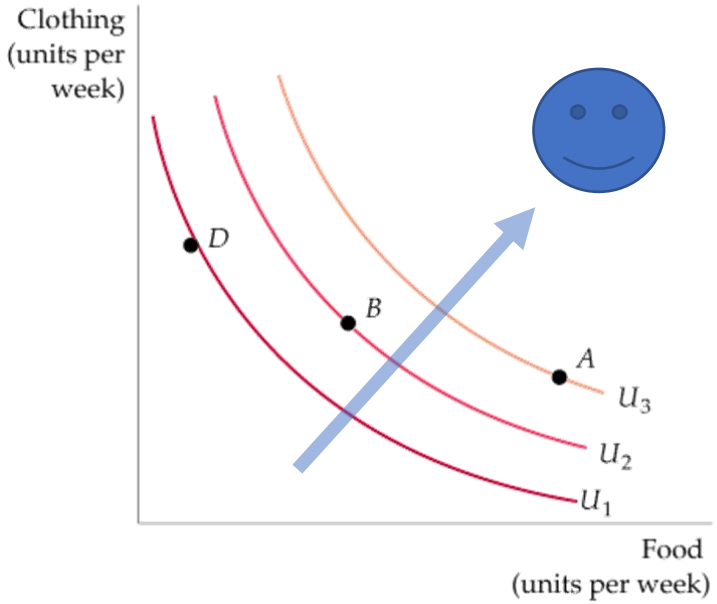
- A is preferred to H



Indifference Maps (describes a consumer's preference)

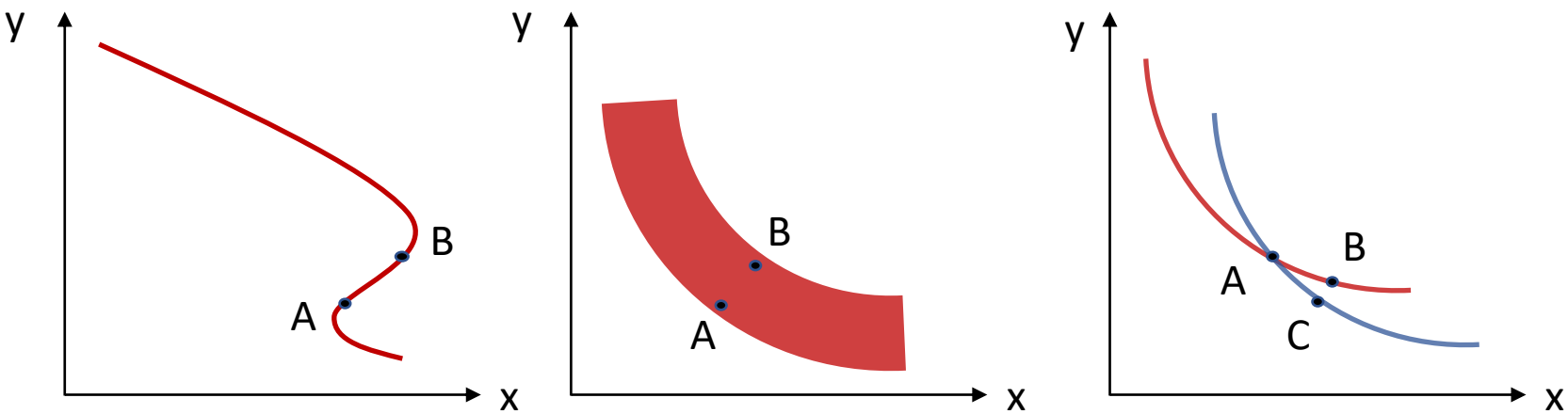
Indifference Map: Graph containing a set of indifference curves showing the market baskets among which a consumer is indifferent.

- Infinitely many curves
- Because more is better, northeast curves are preferred to southwest ones.



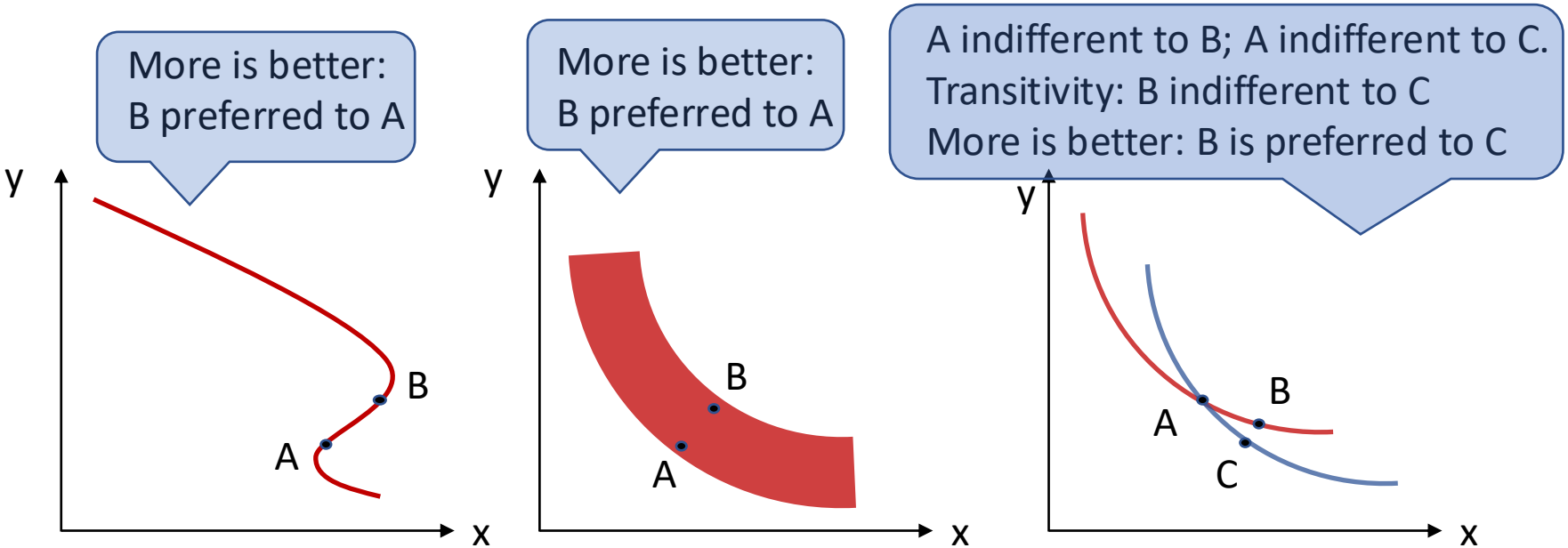
Indifference Maps

- **Suppose the three basic assumptions are satisfied.**
- Can indifference curves have an upward sloping part?
- Can indifference curves be thick lines?
- Can indifference curves intersect with each other?



Indifference Maps

- **Suppose the three basic assumptions are satisfied.**
- Can indifference curves have an upward sloping part? No.
- Can indifference curves be thick lines? No.
- Can indifference curves intersect with each other? No.



Violate more is better

Violate more is better

Violate more is better or transitivity

Indifference Maps

- If preferences satisfy the three basic assumptions: completeness, transitivity, and more is better, then indifference curves should be downward sloping thin curves that never intersect with each other.

- There are infinitely many curves on an indifference map, but we typically only draw a few.

- Northeast curves are preferred to southwest ones.

Utility and Utility Function

Sometimes, you are not given the indifference curves/map to describe preferences, instead you are given utility functions.

Utility: Numerical score representing the satisfaction that a consumer gets from a given market basket.

Utility Function: Formula that assigns a level of utility to individual market baskets.

For example:

- $U(x, y) = x + 2y$ (When I consume x unit of Good X and y of Y, $U(x, y)$ is my satisfaction level)

- $U(F, C) = FC$ (When I consume F unit of Food and C unit of clothing, my satisfaction level is $U(F, C)$)

Utility and Utility Function

How to plot the indifference curves when given utility functions?

- Give $U(x, y)$ a value c_1
- Plot all combinations of (x, y) such that $U(x, y) = c_1$
- Given $U(x, y)$ a different value c_2 ……
- ……

Utility and Utility Function

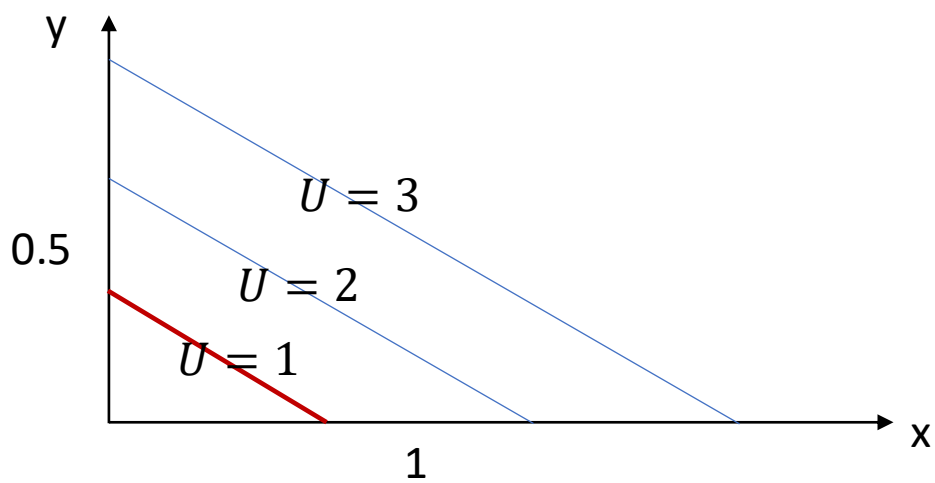
Example: Plot indifference curves of $\boldsymbol{U(x,y) = x + 2y}$

Fix utility level 1 (or any other number). Find all (x, y) such that $x + 2y = 1$.

How to plot all (x, y) such that $x + 2y = 1$?

Solve for y: $y = -0.5x + 0.5$. Then plot and get indifference curve $U(x, y) = 1$.

Repeat for utility levels 2 and 3 etc.



Utility and Utility Function

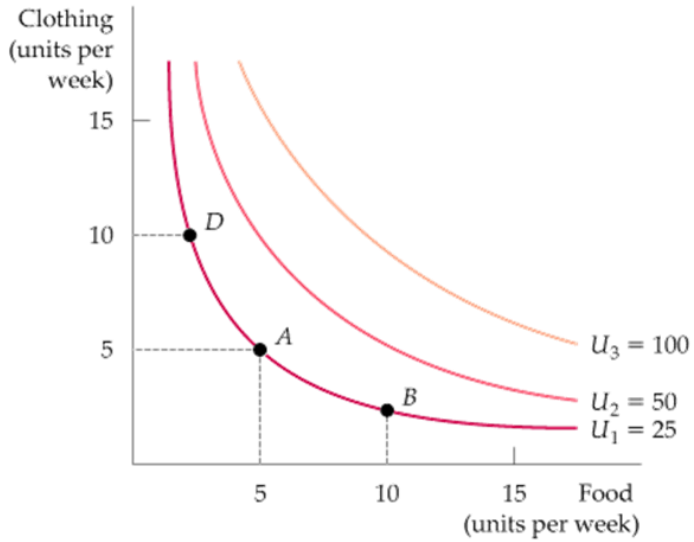
Example: Plot indifference curves of $U(F, C) = FC$

Fix utility level 25 (or any other number). Find all (F, C) such that $FC = 25$.

How to plot all (F, C) such that $FC = 25$?

Solve for C: $C = 25/F$. Plot and get the indifference curve $U(F, C) = 25$.

Repeat for utility level 50 and 100 etc.



Utility and Utility Function

Example: Given $U(F, C) = 4FC^2$. What is the function form of the indifference curve that passes (1, 9)?

Solution:

- Step 1: Find the utility value given by market basket (1, 9).

$$U(1, 9) = 324.$$

- Step 2: Give the utility function the above value.

$$\text{Let } U(F, C) = 4FC^2 = 324.$$

- Step 3: Solve the function of the indifference curve.

Solve for C, we have $C^2 = 81/F$. Hence, $C = 9/F^{0.5}$. It is a downward sloping curve that bends in towards the origin.

You can use online graphing calculators to check: e.g., https://www.desmos.com/calculator.

Utility and Utility Function

Make sure you know how to plot the following two types of indifference curves:

If the utility function is of the form $U(x, y) = ax + by$ with $a, b > 0$, then the indifference curves are downward sloping straight lines.

If the utility function is of the form $U(x, y) = cx^a y^b$ with $a, b, c > 0$, then the indifference curves are downward sloping curves that bend in towards the origin.